

Chapter 5: AI in Education

Digital Education (CS-411)

October 5, 2026

Toolbox of learning mechanisms

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How can AI support these mechanisms?

Just-in-Time Telling

Provide information at the moment it is needed.

Contrasting Cases

Compare carefully designed examples that differ in key ways.

Undoing

Expose and challenge existing misconceptions or incorrect knowledge.

Self-Explanation

Prompt learners to explain their reasoning and reflect on it.

Reward

Provide contingent rewards to motivate and reinforce learning.

Worked Example

Show a complete solution to a problem, often with explanations of each step.

Feedback

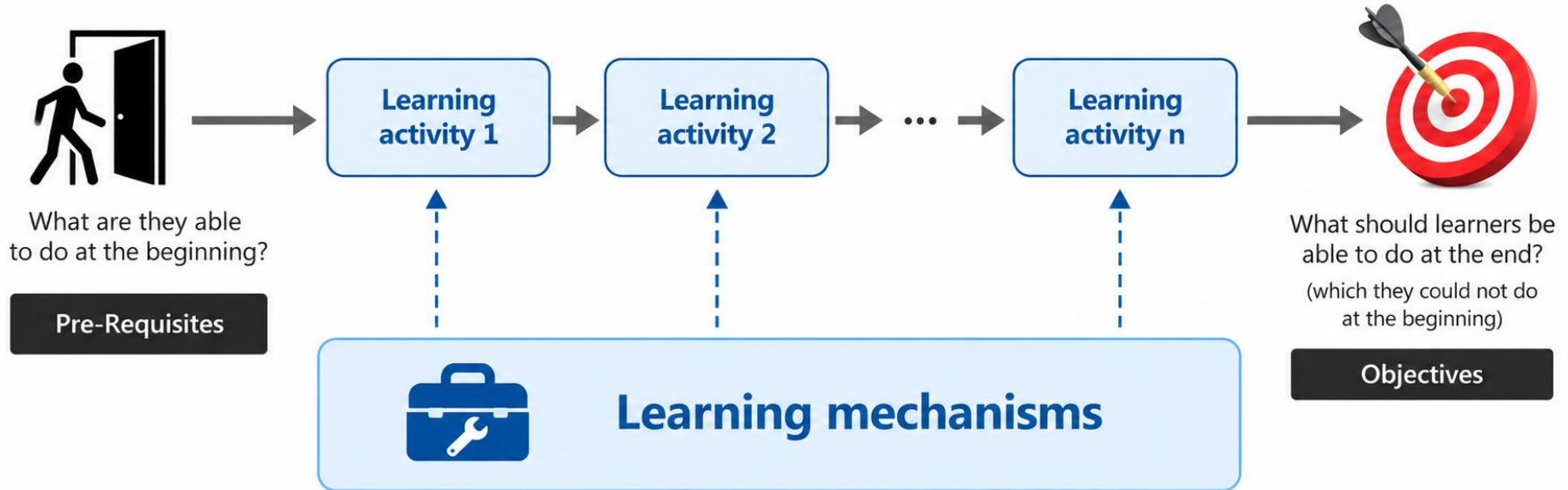
Provide information about performance and how to improve.

Deliberate Practice

Provide focused practice on a specific skill, with opportunities for repetition and refinement.

AI in education $\neq$ AI tutoring

AI is not a learning mechanism



History of AI in education



Computer-Assisted Instruction (CAI)

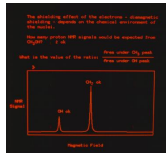
Intelligent Tutoring Systems (ITS)

Conversational Tutoring Systems (CTS)

Learning Analytics (LA)

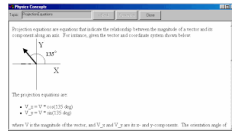


Cognitive Tutor



PLATO

Static Branching



Andes

Symbolic Adaptivity



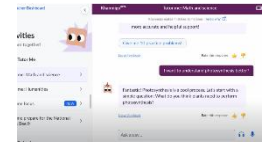
AutoTutor

Machine Learning



Squirrel AI

Deep Learning

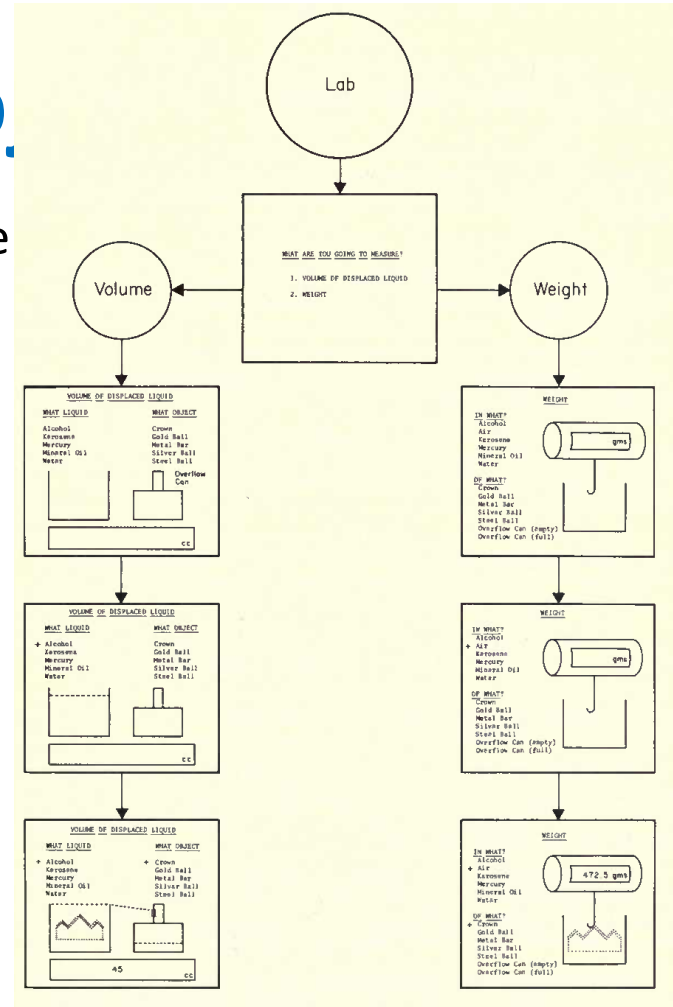
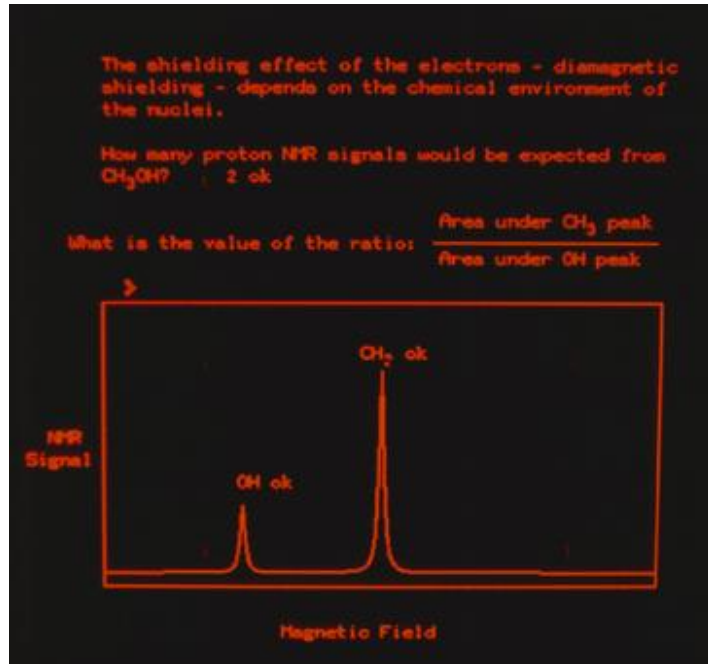


Khanmigo

Generative AI

The PLATO Pro.

- Programmed Logic for Automatic Teaching Operations



Symbolic Adaptivity: Andes

ANDES Physics Workbench - [dt5a.fbd]

File Edit Diagram Variable View Help

A 2000-kg car in neutral at the top of a 20.0 deg inclined driveway 20.0 m long slips its parking brake and rolls down.

If we ignore friction and drag, what would the magnitude of the velocity of the car be when it hits the garage door?

Answer:

Diagram: A car is shown at the top of a 20.0 m inclined driveway at a 20.0 degree angle. A coordinate system is shown with +Y pointing up the incline, +X pointing down the incline, and a green arrow labeled 'd' pointing down the incline. A green arrow labeled 'Fw' points vertically downwards from the car.

Variables:

Name	Definition
T0	car starts rolling
T1	car hits garage door
x	axis
mc	mass of car
d	magnitude of the Displacement of
Fw	magnitude of the Weight Force on

1. $mc = 2000 \text{ kg}$

2. $d = 20.0 \text{ m}$

3. $Fw_y = mc * g$

4.

5.

6.

7.

8.

9.

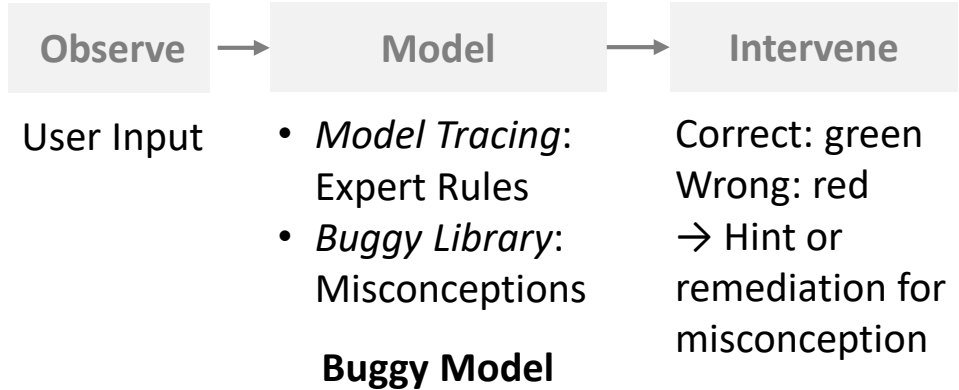
10.

T: Now that you have stated all of the given information, you should start on the major principles. What quantity is the problem seeking?

S: The magnitude of the instantaneous Velocity of car at time T1

T: Yep. What is the first principle application that you would like to work on? Hint: this principle application will usually be one that mentions the sought quantity explicitly. Therefore it's equation may contain the sought quantity that the problem seeks.

270 degrees



Probabilistic Model: Cognitive Tutor

The screenshot displays the Cognitive Tutor interface with four main components:

- Scenario:** A person works on a commission basis, selling magazine subscriptions. For each subscription he sells, he earns five dollars. To become a representative of the company, he first had to pay the company a fee of \$100 to take its training course. His profit equals the money he gets from sold subscriptions minus his fee for the training course.
 - If he sells 80 subscriptions, how much profit will he make?
 - How much profit does he make if he sells one hundred subscriptions?
 - If 56 subscriptions are sold, by how much will he profit?

Please graph the profit as a function of the subscriptions sold, and then use the graph to answer questions 4 and 5 below.

 - If he sells forty subscriptions, how much profit will he make?
 - How many subscriptions must he sell to make a profit of \$150?

To write an expression, define a variable for the number of magazine subscriptions sold and use this variable to write a rule for the salesman's profit.

Problem 5H1139
- Worksheet:** A table for recording data.

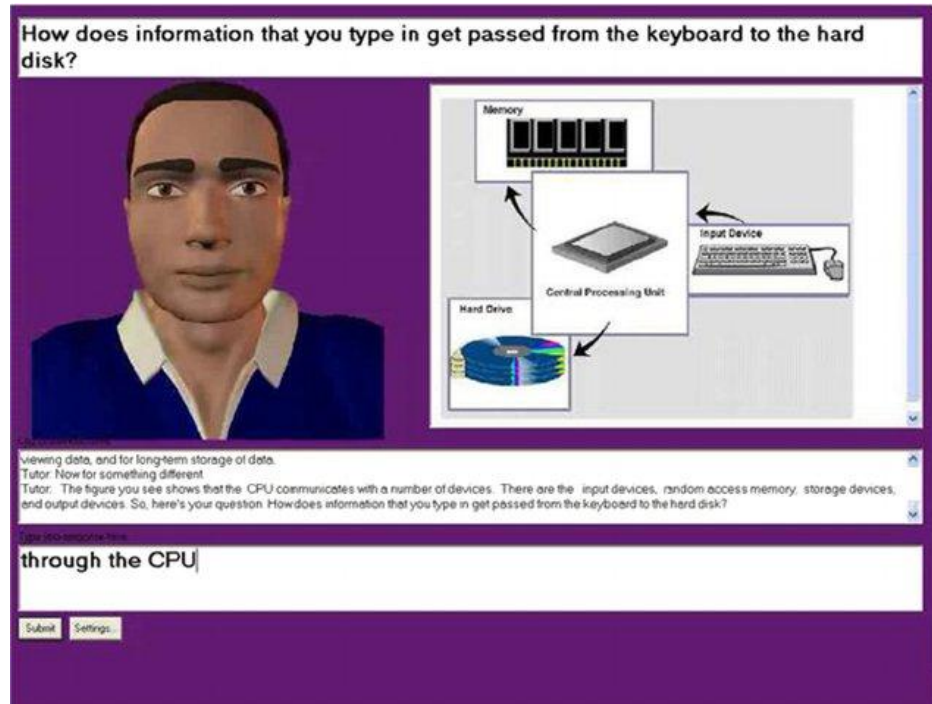
Quantity Name	NUMBER OF SUBSCRIPTIONS SOLD	PROFIT
Unit	SUBSCRIPTIONS	DOLLARS
Question 1		
Question 2		
Question 3		
Question 4		
Question 5		
Expression		
- Skills:** A bar chart titled "steve ritter's skills" showing proficiency levels for various math skills. The skills listed are: identifying units, entering a given, Write expression, positive slope, Find V, positive slope, Using large numbers, Using simple numbers, Correctly placing points, Changing axis bounds, and Changing axis intervals. A yellow bar highlights the "skills" category.
- Grapher:** A window for plotting a graph. It includes a grid with X and Y axes ranging from 0 to 10. A table for axis bounds is shown:

	Lower Bound	Upper Bound	Interval
X Bounds	0.0	10.0	1.0
Y Bounds	0.0	10.0	1.0

The graph is titled "Point Plotting".

- *Domain Model:* skills for math word problem-solving
- *Student Model:* probabilistic model
- *Pedagogical model:* mastery learning, advance when skill is learned

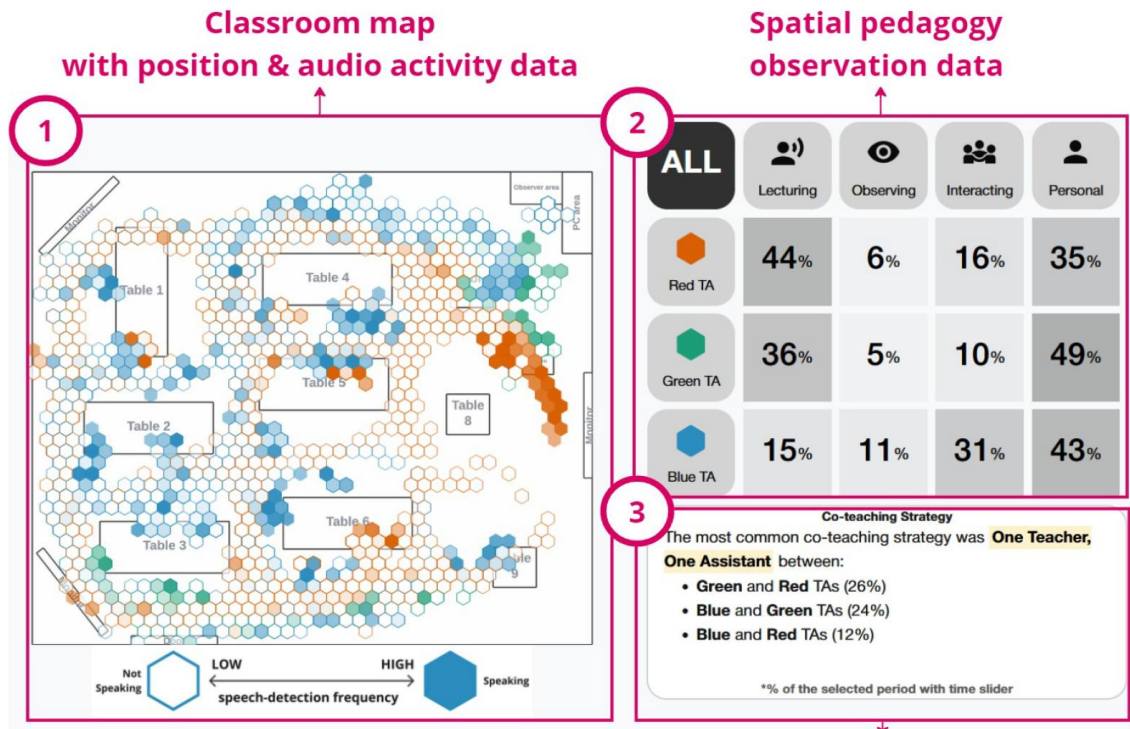
ML-Based: AutoTutor



Definition

Learning analytics is the *collection, analysis, interpretation and communication of data about learners* and their learning that provides theoretically relevant and actionable *insights to enhance learning and teaching*.

Student-Facing Dashboard: TeamTeachingViz



Co-teaching strategy descriptive data

R. Alfredo, P. Mejia-Domenzain, V. Echeverria, D. Rahayu, L. Zhao, H. Alajlan, Z. Swiecki, T. Käser, D. Gašević, and R. Martinez-Maldonado (2025). TeamTeachingViz: Benefits, Challenges, and Ethical Considerations of Using a Multimodal Analytics Dashboard to Support Team Teaching Reflection. *Proceedings of LAK*

What can AI do to support learning?

GENERATE

Create new content

problems · examples ·
explanations · feedback ·
scenarios

ANALYZE

Make sense of
learner activity

answers · solutions ·
explanations ·
interaction traces

ADAPT

Change what is provided
based on context or
learner needs

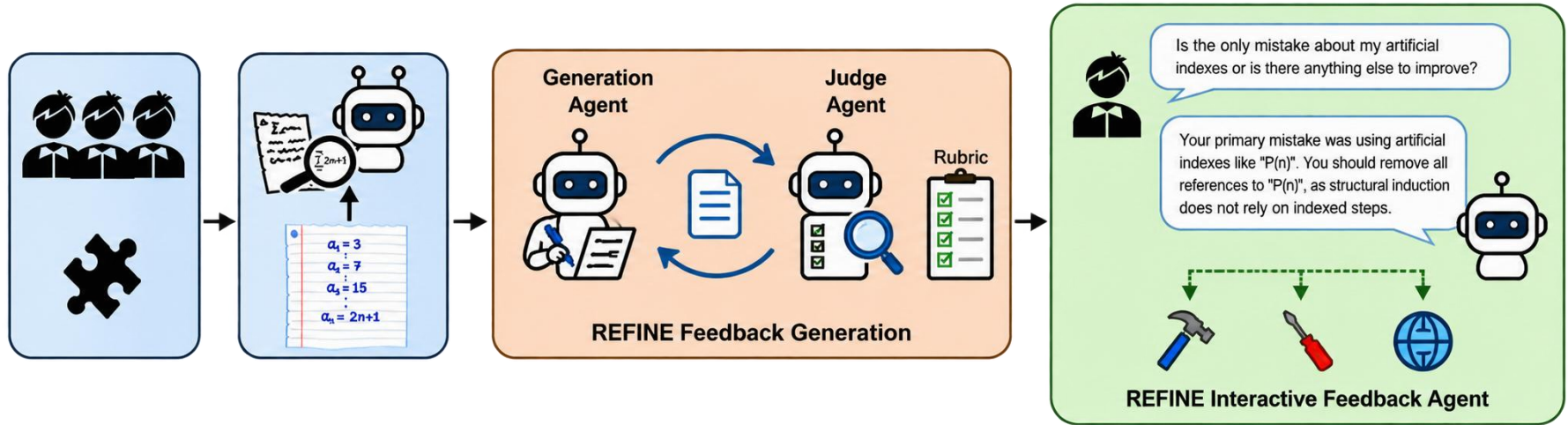
difficulty · content ·
representation ·
next activity

INTERACT

Participate dynamically
in the learning activity

dialogue · questioning ·
simulation · scaffolding

REFINE – Interactive feedback



ANALYZE

GENERATE

INTERACT

FEEDBACK

PharmaSim – Supporting diagnostic reasoning



ANALYZE

ADAPT

INTERACT

SCAFFOLDING

Güres, F. B., Nazaretsky, T., Neshaei, S. P., & Käser, T. (2026). Structuring versus Problematising: How LLM-based Agents Scaffold Learning in Diagnostic Reasoning, Proc. LAK, <https://doi.org/10.1145/3785022.3785105>

Scaffolding

Core idea:

Learners receive temporary support that enables them to engage in a task they could not yet manage independently.

Why it works:

Scaffolding makes complex tasks manageable while keeping learners actively engaged in the learning process. Support can be adjusted as learners develop greater competence.

Two forms:

Structuring: helps learners manage complexity, focus attention, and organize the task.

Problematizing: prompts learners to articulate their thinking, notice gaps, and reconsider their reasoning.

PharmaSim – Supporting diagnostic reasoning



ANALYZE

ADAPT

INTERACT

GENERATE

SCAFFOLDING

Güres, F. B., Nazaretsky, T., Neshaei, S. P., & Käser, T. (2026). Structuring versus Problematising: How LLM-based Agents Scaffold Learning in Diagnostic Reasoning, Proc. LAK, <https://doi.org/10.1145/3785022.3785105>

Relex – Supporting procedural writing

RELEX - Recipe Learning through Examples

A simple tool to help you improve your recipe writing skills.

The interface is divided into several sections:

- Reflection Space:** Contains two questions for reflection on a recipe example. The first question asks for interesting, useful, convincing, and which not? The second question asks for deficiencies compared to the example recipe. A 'Save' button is present.
- Text Editor:** A text area for writing a recipe. Below it is a 'Text Editor' label and a prompt to write a recipe here, then click on the 'Analyze' button to receive personalized tips shown in an example recipe. A 'Save' button is present.
- Personal Dashboard:** Shows a recipe chosen based on the user's text, exemplifying some improvements they can make. It includes a 'Details' link.
- Tips on the Specificity of Ingredients:** A green box with a question mark icon. It contains a tip: 'Indicate the form of olives.' and another tip: 'Indicate the type of pepper.'
- Tips on the Specificity of Steps:** A blue box with a question mark icon. It contains a tip: 'Specify the intensity of cooking heat.'
- Tips on Structure:** A yellow box with a question mark icon. It contains three tips: 'Write the ingredients and the steps in separate sections.', 'List each ingredient separately.', and 'Enumerate the steps.'
- Tips for Clarity of Steps:** A teal box with a question mark icon. It contains two tips: 'Describe how a mixture should look or feel at a certain stage.' and 'Describe until when to do the actions.'

The interface also includes a 'Help' button and a 'Save and Reset' button.

ANALYZE

ADAPT

WORKED EXAMPLE

SELF-EXPLANATION

Gelex – Supporting procedural writing

Recipe : Coconut, some dates, apricots, and cranberries. A few cashews, almonds—however much you want. Get some flour. Grease some pans. Mix everything in a big bowl, starting with the coconut and nuts. Add the flour. Pour milk into the mix and stir. Throw the mixture into some pans. Bake in the pre-heated oven and let it cool for a bit. Take cakes from pans. After cooling, they're fine to eat.

Get improved recipe

Improved Recipe

Title: Coconut and Nut Fruit Cake

Ingredients:

- 1 cup shredded coconut (unsweetened)
- 1 cup dates, pitted and chopped
- 1 cup dried apricots, chopped
- 1 cup dried cranberries
- 1/2 cup cashews, roughly chopped
- 1/2 cup almonds, roughly chopped
- 2 cups all-purpose flour
- 1 cup milk
- Butter or oil for greasing pans
- Pinch of salt

Steps:

1. Preheat your oven to 350°F (175°C).
2. Grease two 8-inch round cake pans with butter or oil to prevent sticking.
3. In a large mixing bowl, combine the shredded coconut, chopped dates, apricots, cranberries, cashews, and almonds.
4. Add the all-purpose flour to the bowl and mix well to ensure all the ingredients are evenly coated.

Write a small explanation, at least 5 characters

A section title for steps is added to clearly separate the preparation instructions from the list of ingredients.

☒ See a suggested explanation

A section title for steps is added to clearly separate the preparation instructions from the list of ingredients.



ANALYZE

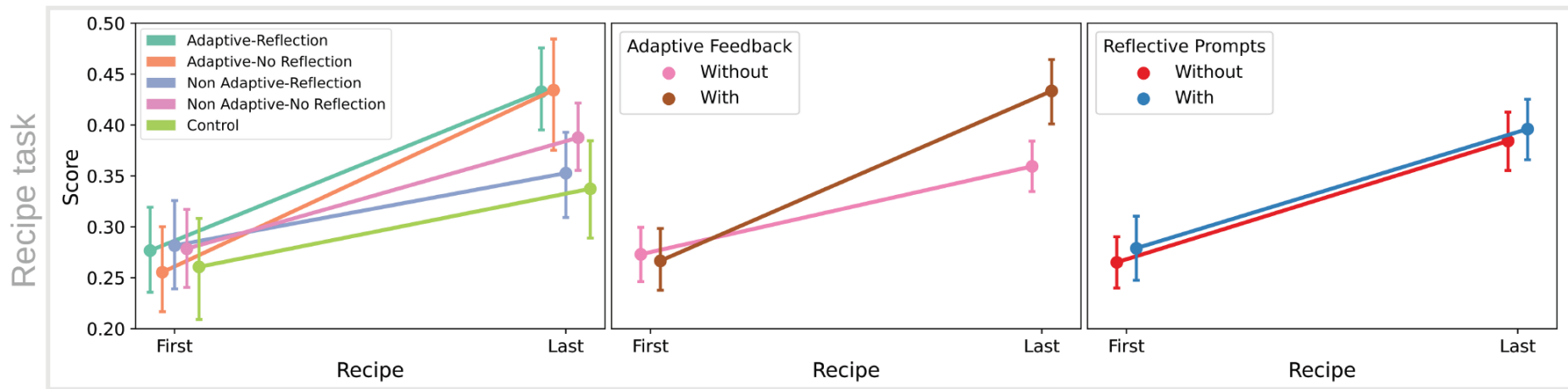
GENERATE

ADAPT

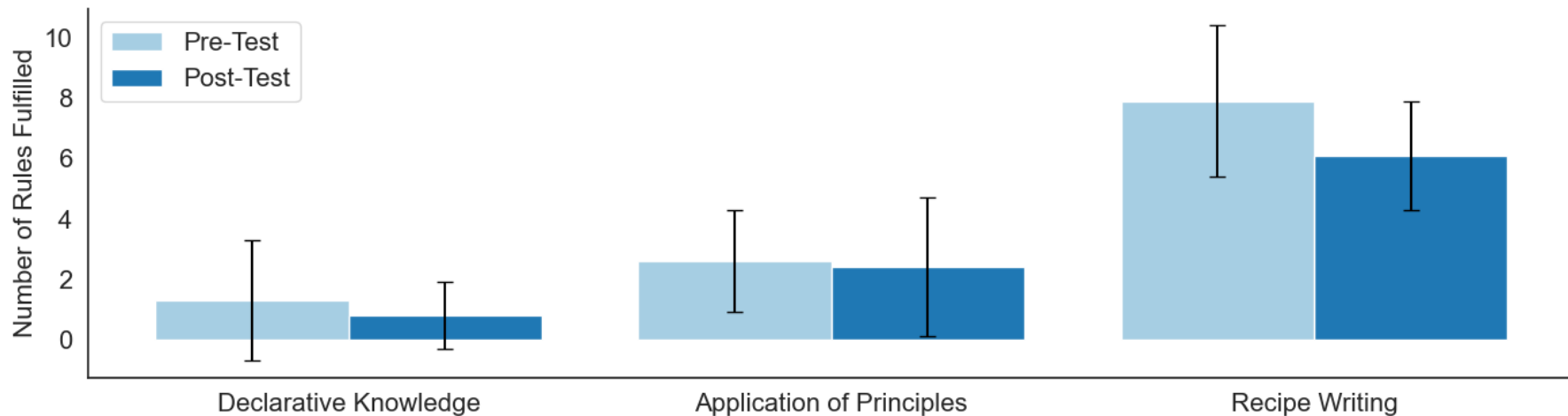
WORKED EXAMPLE

SELF-EXPLANATION

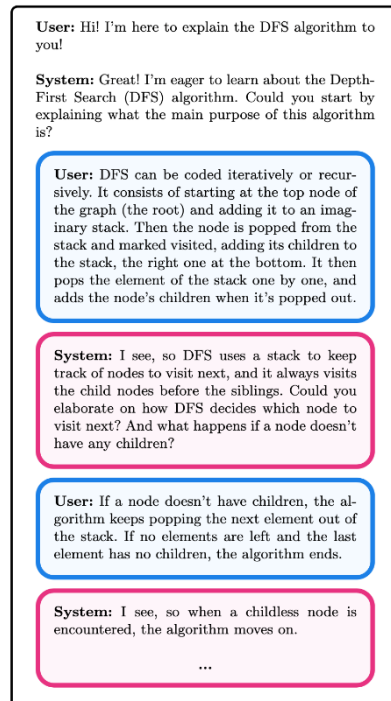
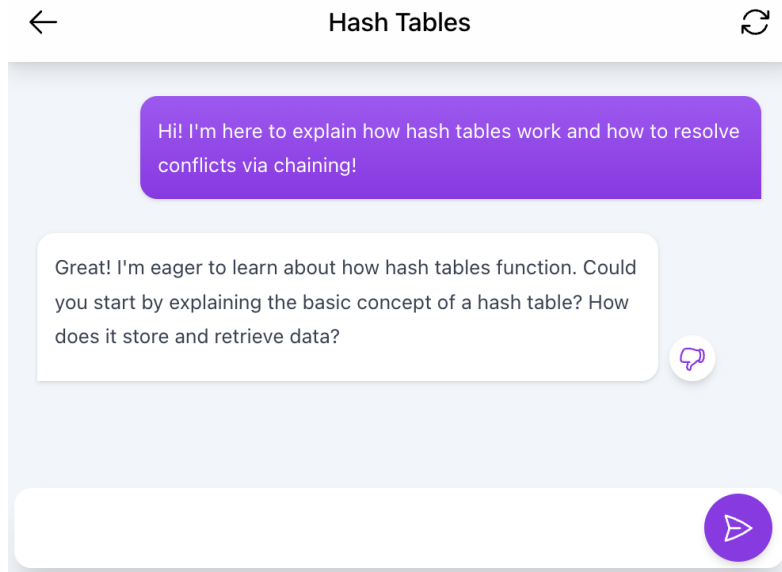
Relex – Results



Gelex – Results



Explicue



INTERACT

TEACHING

Teaching

Core mechanism:

Teaching improves the teacher's own knowledge.

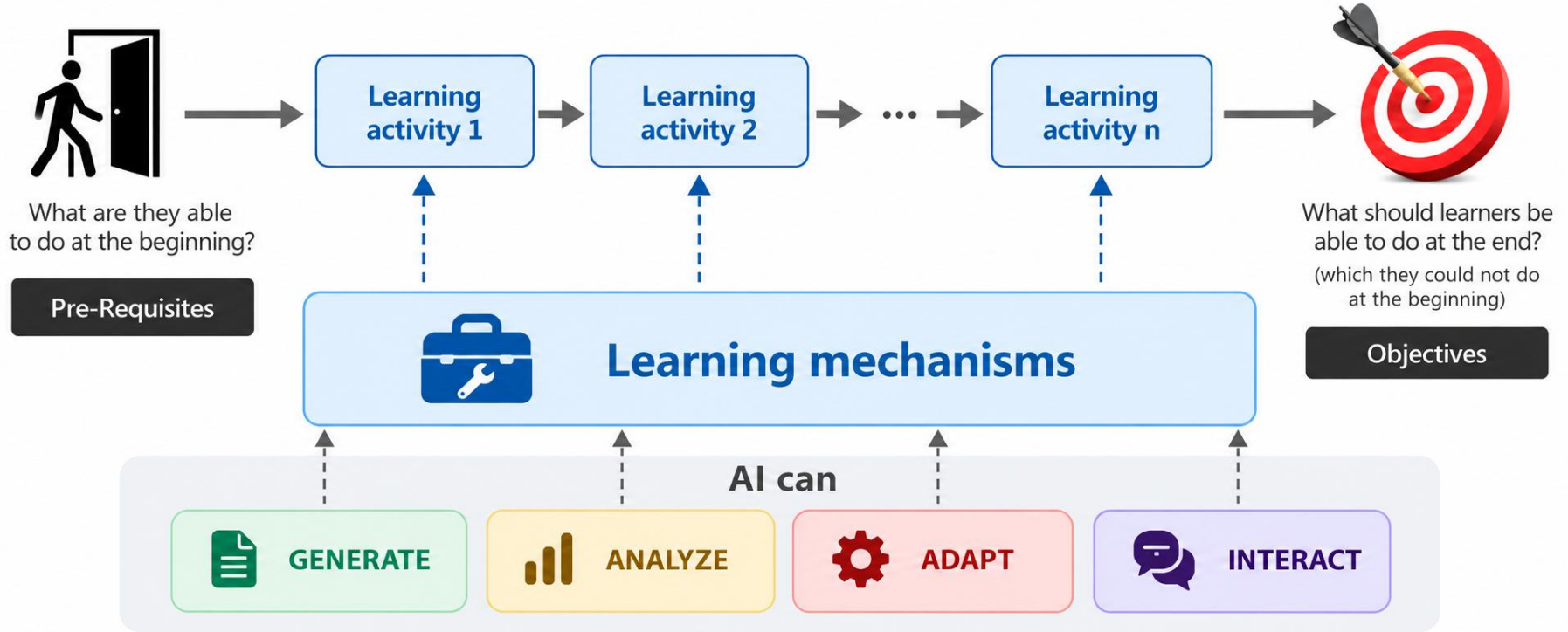
Why it works:

Teaching requires learners to retrieve, organize, and explain their knowledge. Questions from the learner can expose gaps and prompt further elaboration.

Example:

Two learners take turns teaching each other a concept they have just learned.

AI can support learning mechanisms



Design an AI-supported learning activity

- 1) Take one learning activity already planned for your project
Identify the **learning objective** and intended **learning mechanism(s)**.
- 2) Design or prototype AI support
Choose what AI should **GENERATE, ANALYZE, ADAPT, or INTERACT** with.
Design the interaction or build a small proof of concept.
- 3) Experience each other's designs
What is the learner supposed to do?
Which learning mechanism do you experience?
What is the AI doing?
Is AI supporting the learner's productive effort—or replacing it?